OOP Analysis Report – Pillow Library

Library Name: Pillow (PIL Fork)

Group Member	Student ID
Hammad Latif	F25BDATS1M02049
Mudassir Jabbar	F25BDATS1M02052
Javeria Ahmad	F25BDATS1M02074

Date: 07 May 2026

Library Overview

Pillow is a popular Python library used for image processing and image manipulation. It is the modern version of the old Python Imaging Library (PIL).

What Pillow Can Do

- Open images
- Resize images
- Crop images
- Rotate images
- Apply filters and effects
- Draw text and shapes
- Convert image formats

Who Uses Pillow?

Pillow is widely used by Python developers, data scientists, AI engineers, web developers, and automation projects for image handling tasks.

How to Install Pillow

```
Use the following command:

pip install pillow
```

Basic Pillow Example

```
from PIL import Image
```

```
img = Image.open("photo.jpg")
img.show()
img.resize((300,300))
img.save("new_photo.jpg")
```

OOP Concepts Used in Pillow

1. Classes and Objects
2. Encapsulation
3. Inheritance
4. Polymorphism
5. Abstraction

Classes and Objects

```
from PIL import Image img =

Image.open("photo.jpg") Here, Image is a

class and img is an object.
```

Encapsulation

```
img.rotate(90) The user calls methods without knowing internal

implementation.
```

Inheritance

Different filter classes inherit features from parent classes for code reuse and better organization.

Polymorphism

Image.open("a.png")
Image.open("b.jpg")

The same method works with different image formats.

Abstraction

```
img.blur() Complex image processing is hidden
from the user.
```

Advantages of Pillow

• Easy to learn
• Beginner friendly
• Fast image processing
• Supports many formats
• Good documentation
• Works with NumPy and OpenCV

Conclusion

Pillow is an excellent Python library for image processing and demonstrates important OOP concepts like encapsulation, inheritance, polymorphism, and abstraction.